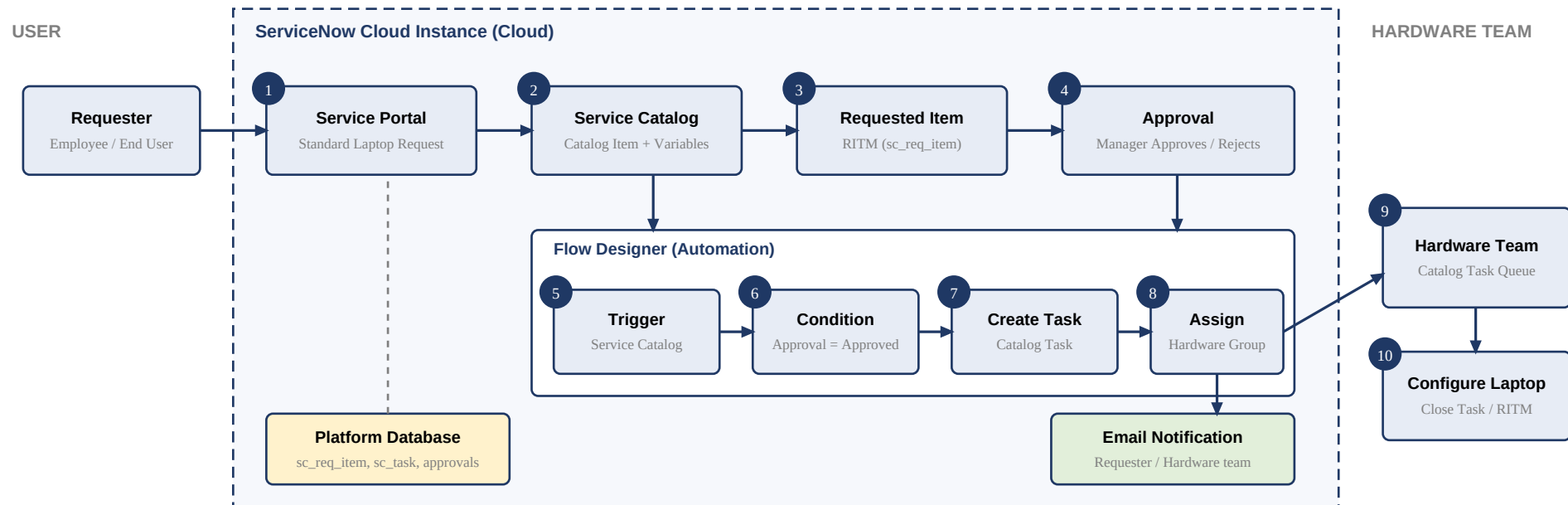


Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	03 October 2026
Team ID	6aba11b55015c35b98631d18
Project Name	Standard Laptop Procurement Automation (ServiceNow)
Maximum Marks	4 Marks

Technical Architecture



Flow: 1 User opens Service Portal, 2 selects Standard Laptop catalog item, 3 RITM is created, 4 manager approval, 5 Flow Designer triggers, 6 checks approval is Approved, 7 creates Catalog Task, 8 assigns to Hardware group, 9 team sees task in queue, 10 configures laptop and closes task.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Requester submits the Standard Laptop request; hardware team works tasks	ServiceNow Service Portal (HTML, CSS, JavaScript, AngularJS widgets)
2	Application Logic-1	Catalog item with variables; creates Requested Item (RITM) on submission	ServiceNow Service Catalog
3	Application Logic-2	Approval of the request by the manager; approve / reject	ServiceNow Approvals (Flow Designer "Ask for Approval")
4	Application Logic-3	Automation: on approval, create Catalog Task and assign to Hardware group	ServiceNow Flow Designer (trigger, condition, Create Catalog Task action)
5	Database	Stores requests, tasks, approvals and groups (sc_req_item, sc_task, sysapproval_approver, sys_user_group)	ServiceNow relational database tables
6	Cloud Database	Database service hosted with the cloud instance	ServiceNow platform database (cloud-hosted)
7	File Storage	Attachments on requests and tasks (if any)	ServiceNow attachment storage (sys_attachment)
8	External API-1	Email notifications to requester and Hardware team	ServiceNow Email Notifications (SMTP)
9	External API-2	Not used in this project	N/A
10	Machine Learning Model	Not applicable	N/A
11	Infrastructure (Server / Cloud)	Application built and deployed on the ServiceNow instance	Local Server Configuration: None (browser access only) Cloud Server Configuration: ServiceNow cloud instance; update set to move between instances

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source libraries used by the portal front end	AngularJS, Bootstrap (via Service Portal)
2	Security Implementations	Role-based access to catalog item, approvals and tasks; encrypted access	ACLs, roles and groups, HTTPS/TLS, authenticated login (SSO optional)
3	Scalable Architecture	Reusable, low-code flow on a multi-tenant cloud platform; new items can reuse the same flow pattern	ServiceNow platform, Flow Designer (subflows)
4	Availability	Hosted on ServiceNow-managed cloud infrastructure; no self-managed servers	ServiceNow cloud hosting
5	Performance	Automation runs on approval event, removing manual task creation; indexed table lookups	Flow Designer event-driven execution, platform database indexing